

Chapter-06

1. Transformation

Definition

A **transformation** is the process of changing the position, orientation, or size of an object without changing its basic identity. In Computer Graphics, transformations are used to:

- Move objects
- Rotate objects
- Resize objects
- Create animation
- Change coordinate systems

The three basic geometric transformations are:

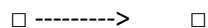
1. Translation
2. Rotation
3. Scaling

Types of Transformations

1. Translation

Moves an object from one position to another without changing its size or orientation. Example:

Before After



2. Rotation

Rotation is a geometric transformation that turns an object by a specified angle (clockwise or counterclockwise) about a fixed point, called the pivot point or center of rotation, while preserving the object's size, shape, and distances between all points. Example:



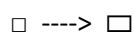
Rotate 45°

3. Scaling

Changes the size of an object.

- Enlarging
- Shrinking

Example:



Exam Definition

A geometric transformation is an operation that changes the position, orientation, or size of an object using mathematical equations.

2. Translation

Definition

Translation moves every point of an object by the same distance in a specified direction.

The object:

- Does NOT rotate
- Does NOT change shape
- Does NOT change size

Only its location changes.

Translation as a Rigid Body Transformation

Definition

A Rigid Body Transformation is a transformation in which

- Shape remains unchanged
- Size remains unchanged
- Angles remain unchanged
- Distance between any two points remains unchanged

Translation satisfies all these conditions. Therefore,

Translation is a Rigid Body Transformation.

Properties

- ✓ Shape preserved
- ✓ Size preserved
- ✓ Angles preserved
- ✓ Parallel lines remain parallel
- ✓ Only position changes

Translation Equation

Suppose

Original point

$$P(x, y)$$

Translation vector

$$T(t_x, t_y)$$

After translation

$$P'(x', y')$$

Then

$$\begin{aligned}x' &= x + t_x \\ y' &= y + t_y\end{aligned}$$

where

- t_x = movement along X-axis
- t_y = movement along Y-axis

These equations define the translation of a point in 2D.

Matrix Form

Translation can be written as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix}$$

Using homogeneous coordinates:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Derivation (Proof) of Translation Equation

Suppose

Original point

$$P(x, y)$$

The object moves

- t_x units horizontally

- t_y units vertically

New x-coordinate

$$x' = x + t_x$$

because we simply add the horizontal displacement. Similarly, New y-coordinate

$$y' = y + t_y$$

Hence,

$$\boxed{x' = x + t_x, y' = y + t_y}$$

This is the translation equation.

Example

Original point

$$(3, 4)$$

Translation vector

$$(5, -2)$$

Then

$$x' = 3 + 5 = 8$$

$$y' = 4 - 2 = 2$$

New point

$$(8, 2)$$

3. Rotation

Rotation is a geometric transformation that turns an object by a specified angle (clockwise or counterclockwise) about a fixed point, called the pivot point or center of rotation, while preserving the object's size, shape, and distances between all points. The object:

- Changes orientation
- Keeps the same shape
- Keeps the same size

Positive angle → Counterclockwise , Negative angle → Clockwise

Characteristics

- Pivot point remains fixed.
- Distance from the pivot remains constant.
- Shape does not change.
- Size does not change.
- Rotation is also a **Rigid Body Transformation**.

Rotation Equation

For rotation about the **origin** by angle θ :

$$x' = x \cos \theta - y \sin \theta$$

$$y' = x \sin \theta + y \cos \theta$$

These are the standard 2D rotation equations.

Matrix Form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Homogeneous form:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Derivation (Proof) of Rotation Equation

Suppose

Original point

$$P(x, y)$$

Using polar coordinates,

$$x = r \cos \phi$$

$$y = r \sin \phi$$

After rotation by angle θ ,

New angle

$$\phi + \theta$$

Using trigonometric identities,

$$x' = r \cos(\phi + \theta)$$

Apply

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Therefore,

$$x' = r(\cos \phi \cos \theta - \sin \phi \sin \theta)$$

Since

$$x = r \cos \phi$$

and

$$y = r \sin \phi$$

Substitute,

$$x' = x \cos \theta - y \sin \theta$$

Similarly,

$$y' = r \sin(\phi + \theta)$$

Apply

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

Substitute,

$$y' = x \sin \theta + y \cos \theta$$

Hence,

$$\boxed{x' = x \cos \theta - y \sin \theta}$$

$$\boxed{y' = x \sin \theta + y \cos \theta}$$

This is the derivation commonly expected in exams.

Example

Rotate

$$(1, 0)$$

through

$$90^\circ$$

Since

$$\cos 90^\circ = 0, \sin 90^\circ = 1$$

Then

$$x' = 1(0) - 0(1) = 0$$

$$y' = 1(1) + 0(0) = 1$$

Therefore,

$$(1, 0) \rightarrow (0, 1)$$

Important Properties of the Rotation Matrix

- A rotation matrix is an orthogonal matrix.
- Its rows and columns are perpendicular (90°) to each other, and each has a length of 1 (unit vector).
- The inverse (undo) of a rotation matrix is simply its transpose (swap rows and columns). In 2D, this is the same as changing the sign of the sine ($\sin$) terms.
- Because of this property, we do not need to calculate a complicated inverse matrix, making calculations faster and easier.
- This property is very useful in computer graphics because it helps us easily rotate objects back to their original position or create a rotation matrix from known directions.

Exam

Tip:

Inverse of Rotation Matrix = Transpose of Rotation Matrix

$$R^{-1} = R^T$$

4. Scaling

Definition: Scaling is the geometric transformation that changes the size of an object.

Scaling can:

- Enlarge an object
- Reduce (shrink) an object

Unlike translation and rotation, scaling changes the dimensions of the object (unless the scale factor is 1).

Scale Factors

Let

- S_x = Scaling factor along x-axis
- S_y = Scaling factor along y-axis

Interpretation:

- $S > 1 \rightarrow$ Enlargement
- $0 < S < 1 \rightarrow$ Reduction
- $S = 1 \rightarrow$ No change
- $S < 0 \rightarrow$ Scaling with reflection

Scaling Equation

Suppose

Original point

$$P(x, y)$$

After scaling,

$$P'(x', y')$$

Then,

$$\boxed{x' = S_x x}$$

$$\boxed{y' = S_y y}$$

These equations scale the point relative to the **origin**.

Matrix Form

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Uniform Scaling

Definition

If

$$S_x = S_y = S$$

then the object is enlarged or reduced equally in every direction. Shape remains the same.
Only size changes.

Example

Scale factor

$$S_x = S_y = 2$$

Original point

$$(3, 4)$$

New point

$$(6, 8)$$

The object becomes **twice as large**.

Properties

- ✓ Shape preserved
- ✓ Angles preserved
- ✓ Aspect ratio preserved
- ✗ Size changes

Differential (Non-uniform) Scaling

Definition

If

$$S_x \neq S_y$$

then different scaling factors are applied along x and y directions.
The object becomes stretched or compressed.

Example

$$S_x = 2, S_y = 1$$

Original

□

After scaling

□

Width doubles but height remains unchanged.

Properties

- ✓ Size changes
- ✓ Shape changes
- ✓ Aspect ratio changes

Comparison

Uniform Scaling	Differential Scaling
$S_x = S_y$	$S_x \neq S_y$
Shape preserved	Shape changes
Aspect ratio preserved	Aspect ratio changes
Example: square → bigger square	Example: square → rectangle

Example

Original point

$$(2, 5)$$

Scale factors

$$S_x = 3, S_y = 2$$

Then

$$x' = 3(2) = 6$$

$$y' = 2(5) = 10$$

New point

$$(6, 10)$$

5. Composite Transformation

A Composite Transformation is obtained by applying two or more transformations one after another. Instead of applying transformations separately, we combine them into one composite transformation matrix, which is computationally efficient.

Why Composite Transformation?

Advantages:

- Faster computation
- One matrix multiplication
- Used in animation
- Used in CAD software
- Used in games

Important Rule

Transformation order is important. Matrix multiplication is not commutative.

That means,

$$A \times B \neq B \times A$$

For example,

Translate → Rotate

is NOT the same as

Rotate → Translate.

Translation + Rotation

Sequence

Object

↓

Translate

↓

Rotate

Composite Matrix

$$M = R \times T$$

because the rightmost matrix acts first on the point (column-vector convention).

Uses

- Animation

- Robot movement
- Camera motion

Rotation + Scaling

Sequence

Rotate

↓

Scale

Composite Matrix

$$M = S \times R$$

Uses

- Zooming rotated objects
- Image editing
- CAD

Translation + Scaling

Sequence

Translate

↓

Scale

Composite Matrix

$$M = S \times T$$

Uses

- Resize after moving
- Logo manipulation

Translation + Rotation + Scaling

Sequence

Translate

↓

Rotate

↓

Scale

Composite Matrix

$$M = S \times R \times T$$

Apply to point

$$P' = M \times P$$

Exam Note: Always remember:

Rightmost transformation is applied first.

Example: Suppose

Translate

$$(3, 2)$$

Rotate

$$90^\circ$$

Scale

$$2$$

Instead of applying three separate equations, Multiply

$$M = SRT$$

Then

$$P' = MP$$

This is why composite transformations are preferred.

6. General Pivot-Point Rotation

Definition

Normally, Rotation occurs about the origin. But in practice, Objects often rotate about any arbitrary point. That point is called the Pivot Point.

Suppose

Pivot

$$(x_r, y_r)$$

Rotation angle

$$\theta$$

Steps of Pivot Rotation

Step 1

Translate pivot to the origin

$$T(-x_r, -y_r)$$

Step 2

Rotate about origin

$$R(\theta)$$

Step 3

Translate back

$$T(x_r, y_r)$$

Composite Matrix

$$M = T(x_r, y_r) R(\theta) T(-x_r, -y_r)$$

This is the standard matrix for rotation about an arbitrary pivot point.

Rotation Equation About Arbitrary Point

The resulting coordinate equations are

$$x' = x_r + (x - x_r)\cos \theta - (y - y_r)\sin \theta$$

$$y' = y_r + (x - x_r)\sin \theta + (y - y_r)\cos \theta$$

These equations are derived by translating the pivot to the origin, rotating, and translating back.

Why Translate First?

Because the standard rotation matrix works only about the origin. Therefore, we first move the pivot to the origin, perform the rotation, and then restore the original position.

Example

Pivot

$$(2, 3)$$

Rotate

$$45^\circ$$

Sequence

Translate (-2,-3)

↓

Rotate 45°

↓

Translate (+2,+3)

7. Fixed-Point Scaling

Definition

Normal scaling occurs relative to the **origin**. Sometimes, we want scaling about a specific point. That point is called the **Fixed Point**. Suppose, Fixed Point

$$(x_f, y_f)$$

Steps

Step 1

Translate fixed point to origin

$$T(-x_f, -y_f)$$

Step 2

Scale

$$S(S_x, S_y)$$

Step 3

Translate back

$$T(x_f, y_f)$$

Composite Matrix

$$M = T(x_f, y_f) S(S_x, S_y) T(-x_f, -y_f)$$

Scaling Equation About Fixed Point

The coordinate equations are

$$x' = x_f + S_x(x - x_f)$$

$$y' = y_f + S_y(y - y_f)$$

Diagram

Object

↓

Move fixed point to origin

↓

Scale

↓

Move back

Why Translate?

The standard scaling matrix works only about the origin.

To scale about another point, we:

1. Move that point to the origin.

2. Apply scaling.
3. Move it back.

8. Other Transformations

A. Reflection

Reflection is a transformation that produces the **mirror image** of an object about a specified line (reflection axis).

Reflection:

- Changes orientation
- Keeps the same size
- Keeps the same shape

Reflection is commonly performed about:

- X-axis
- Y-axis
- Origin
- Line $y = x$
- Line $y = -x$

Reflection about X-axis

Every y-coordinate changes sign.

$$\begin{aligned}x' &= x \\y' &= -y\end{aligned}$$

Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about Y-axis

Every x-coordinate changes sign.

$$\begin{aligned}x' &= -x \\y' &= y\end{aligned}$$

Matrix

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about Origin

Both coordinates change sign.

$$\begin{aligned}x' &= -x \\y' &= -y\end{aligned}$$

Matrix

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about Line $y = x$

Coordinates are exchanged.

$$\begin{aligned}x' &= y \\y' &= x\end{aligned}$$

Matrix

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reflection about Line $y = -x$

Coordinates are exchanged and signs change.

$$x' = -y$$

$$y' = -x$$

Matrix

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Properties

- ✓ Shape preserved
- ✓ Size preserved
- ✓ Mirror image produced
- ✓ Reflection is an affine transformation.

B. Shear

Definition: A Shear Transformation changes the shape of an object by shifting one coordinate direction while keeping the other fixed.

It converts: Rectangle $\rightarrow$ Parallelogram

Shape changes, but parallel lines remain parallel.

Types of Shearing

1. X-Shear

Only x-coordinate changes.

$$x' = x + Sh_x y$$

$$y' = y$$

Matrix

$$\begin{bmatrix} 1 & Sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Y-Shear

Only y-coordinate changes.

$$x' = x$$

$$y' = y + Sh_y x$$

Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Properties

- ✓ Shape changes
- ✓ Size may change

- ✓ Parallel lines remain parallel
- ✓ Angles change

9. Transformation Between Coordinate Systems

Definition

Transformation between coordinate systems means changing the coordinates of an object from one coordinate system to another. For example, an object may be created using its own coordinate system (Local Coordinates), but to display it in a scene we need to convert it to the World Coordinate System.

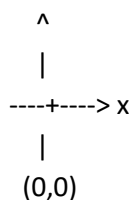
Types

1. Local Coordinate System (Object Coordinates)

- Coordinates are measured relative to the object itself.
- The object's center (or a chosen point) is usually taken as **(0,0)**.

Example:

Car



The car is designed like this:

Wheel = (-2,-1)

Door = (1,0)

Window = (2,1)

These coordinates are inside the car, not in the world.

Think of it as the car carrying its own coordinate system wherever it goes.

2. World Coordinate System

This coordinate system represents the **entire scene**.

Example:

World

Building

Road

Car

(250,120)

Here,

Car position = (250,120)

means the whole car is located at (250,120) in the world.

Local → World Transformation

Usually done by

- Translation
- Rotation

Sometimes Scaling

Steps

Suppose

New coordinate system

Origin

$$(x_0, y_0)$$

Angle

$$\theta$$

Then

Step 1

Translate

Step 2

Rotate

These transformations convert local coordinates into world coordinates.

Applications

- Animation
- Robot movement
- CAD
- Video games
- Object modeling

10. Affine Transformation

An Affine Transformation is a transformation in which every transformed coordinate is a linear function of the original coordinates.

It combines

- Translation
- Rotation
- Scaling
- Reflection
- Shearing

into one mathematical model.

General Affine Equation

$$x' = a_1x + b_1y + t_x$$

$$y' = a_2x + b_2y + t_y$$

where

- a_1, a_2, b_1, b_2 are constants
- t_x, t_y are translation values

Affine Transformation Matrix

$$\begin{bmatrix} a_1 & b_1 & t_x \\ a_2 & b_2 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

Properties

- ✓ Parallel lines remain parallel.
- ✓ Straight lines remain straight.
- ✓ Finite points remain finite.
- ✓ Can represent translation, rotation, scaling, reflection, and shearing.

Examples

Translation

Move object

Rotation

Turn object

Scaling

Resize object

Reflection

Mirror image

Shear

Rectangle → Parallelogram